The Citrus Fruit Borer *(Gymnandrosoma aurantianum):* A Comprehensive Review of Biology, Impact, Management, and Data Resources

I. Executive Summary

The citrus fruit borer, *Gymnandrosoma aurantianum* (Lima, 1927), formerly known as *Ecdytolopha aurantiana*, is a tortricid moth that represents a significant agricultural pest affecting citrus and several other fruit crops throughout Central and South America, and has recently been detected in the United States. The larvae bore into developing fruit where they feed internally on the pulp or seeds, making the fruit unsuitable for consumption or processing and often causing premature fruit drop. Since larval development occurs inside the fruit, infestations are difficult to detect early, allowing populations to grow before control measures can be implemented. In major citrus producing regions such as Brazil, damage caused by the citrus fruit borer has resulted in substantial economic loss each year, highlighting the importance of effective monitoring and management strategies.

Research over the past several decades has improved the understanding of the biology, host range, and ecological behavior of *G. aurantianum*. This pest is highly polyphagous and has been reported from numerous economically important crops including citrus, macadamia, cacao, guava, and lychee. Its relatively short life cycle, combined with favorable tropical conditions, allows for multiple generations per year, contributing to rapid population growth in agricultural systems. As a result, integrated pest management strategies combining pheromone-baited trap monitoring, biological control using parasitoids, and selective insecticides have been developed to reduce crop losses while minimizing environmental impacts. Continued research and international collaboration remain important for improving sustainable control methods and reducing the economic impact of this pest on fruit production systems.

II. Introduction

The citrus fruit borer (CFB) is a moth belonging to the family Tortricidae (order Lepidoptera)[1]. This species is recognized as an important agricultural pest affecting citrus and several other fruit crops throughout Central and South America[2]. The larvae develop inside host fruit where they feed on the pulp or seeds, causing internal damage that can lead to fruit rot, premature fruit drop, and significant economic losses for growers[3]. Since the larvae remain concealed within the fruit during much of their development, infestations are often difficult to detect until visible damage has already occurred[4].

Historically, the species has appeared in scientific literature under several different names, which has caused some confusion in agricultural and entomological studies. Earlier publications commonly referred to the insect as *Ecdytolopha aurantiana*, though later taxonomic revisions reassigned the species to the genus *Gymnandrosoma*[5]. As a result, both names may still appear in older literature, but *Gymnandrosoma aurantianum* is now recognized as the accepted scientific name.

The CFB has a wide geographic distribution throughout tropical and subtropical regions of the Americas, including countries such as Brazil, Argentina, Colombia, Peru, Venezuela, and several nations in Central America and the Caribbean[2]. Favorable environmental conditions in these regions allow the insect to produce multiple generations each year, contributing to rapid population growth in orchards and plantations[6]. In addition to citrus crops, the insect has been reported attacking several other economically important fruit species including macadamia, guava, cacao, peach, lychee, and pomegranate[7]. This broad host range allows this pest to become established across a wide variety of agricultural systems.

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Recent reports have confirmed the detection of the citrus fruit borer in the United States. In 2025, CFB was reported in both Florida and Texas, representing its first known occurrence in the country[12,13]. At this time, the pest is considered transient, meaning it has been detected but is not yet widely established. The presence of the CFB in the United States raises concerns regarding its potential spread and the risk it poses to domestic citrus production systems.

Due to its destructive feeding behavior and ability to infest multiple fruit crops, the CFB has become a major concern for fruit production industries in affected regions. In Brazil, one of the world's largest citrus producers, it has been responsible for substantial economic losses despite intensive management efforts[3,8]. For this reason, extensive research has focused on understanding the biology, ecology, and management of the citrus fruit borer in order to develop effective monitoring strategies and integrated pest management programs.

III. Biology and Identification of Citrus Fruit Borer

Understanding the biology and identifying (morphological) characteristics of the CFB is essential for effective monitoring and management in agricultural systems. In addition to its distinctive life cycle, the citrus fruit borer is also characterized by its broad host range and feeding behavior, which allow it to exploit multiple fruit crops across tropical and subtropical regions[7].

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Life Cycle: Egg, Larve, Pupae and Adult Stages

The CFB undergoes complete metamorphosis consisting of four developmental stages: egg, larva, pupa, and adult [4]. Under favorable environmental conditions, the full life cycle from egg to adult typically lasts approximately 30-40 days, although this duration can vary depending on temperature, humidity, and host plant availability. Because of its relatively short development time, this species is capable of producing several generations per year in tropical and subtropical regions where host fruits are available throughout much of the year[6].

Female moths typically lay eggs directly on the surface of developing or mature fruit[4]. In most cases, a single egg is deposited per fruit, which reduces competition between larvae and ensures that the newly hatched larva has immediate access to a food source[8]. Studies have also shown that females tend to avoid laying eggs on very small fruits, often selecting fruits that have reached a certain stage of development where the larva can successfully penetrate the fruit tissues and obtain sufficient nutrition. After hatching, the larva quickly penetrates the fruit rind and begins feeding internally on the pulp or seeds[4]. This feeding behavior protects the larva from many external predators while allowing development within the fruit.

The larval stage is responsible for most of the crop damage associated with this pest. As larvae feed and tunnel through fruit tissues, they create internal cavities that often lead to fruit rot and premature fruit drop[3]. These feeding tunnels may also allow secondary infections by fungi and bacteria to occur within the fruit[4].

When larval development is complete, the mature larva typically exits the fruit and moves to the soil or plant debris where pupation occurs[4]. The insect forms a cocoon and undergoes metamorphosis into the adult moth. After emerging, adults mate and begin the reproductive cycle again, allowing populations to increase rapidly in favorable environments[6].

Table 1: Key Characteristics of Citrus Fruit Borer Life Stages

Life Stage	Size (approx.)	Color/Appearance	Key Distinguishing Features	Typical Location
Eggs	1-1.3mm	Initially pale white or translucent, darkens slightly before hatching	Oval shaped, usually laid singly on fruit surface	Surface of developing or mature fruit
Larvae (Caterpillar)	16-19mm	Creamy white body with brown head capsule	Bores directly into fruit after hatching; feeds internally on pulp or seeds, produces frass near entry hole	Inside citrus or other host fruit
Pupae	10-12mm	Light brown to dark brown	Encased within a protective cocoon during metamorphosis	Soil, leaf litter, or plant debris beneath host trees

Adult Moth	Wingspan 15-20mm	Grayish-brown forewings with subtle darker markings	Small tortricid moth, active mainly at dusk	Orchard canopy, surrounding vegetation

Host Plants and Feeding Behavior

The CFB is considered a polyphagous species, meaning it is capable of feeding on a wide range of host plants across multiple plant families[7]. Although it is most commonly associated with citrus crops, the insect has been recorded attacking numerous economically important fruit species including macadamia (*Macadamia integrifolia*), cacao (*Theobroma cacao*), guava (*Psidium guajava*), peach (*Prunus persica*), lychee (*Litchi chinensis*), pomegranate (*Punica granatum*), and several other tropical fruit species[7]. This broad host range allows the insect to persist across different agricultural systems and contributes to its importance as a pest in regions where multiple fruit crops are cultivated.

Host plant location is believed to involve a combination of sensory cues, including plant odors, fruit volatiles, and visual characteristics of the host plant[7]. Some researchers have suggested that the insect may respond to general "fruity" chemical signals produced by ripening fruits rather than being specialized for a single plant species. This ability to recognize chemical cues common to many fruits may help explain why the citrus fruit borer is capable of exploiting such a diverse group of host plants.

After hatching, the larva rapidly bores through the fruit rind and begins feeding internally on the pulp or seeds[4]. The entry point is usually marked by a small hole surrounded by frass, which is the waste material produced by the feeding larva[3]. Because feeding occurs within the fruit, larvae are protected from many predators and environmental stresses, which increases their chances of survival.

The nutritional quality and physical characteristics of the host plant can also influence larval development and survival. Research has shown that factors such as nitrogen content, plant defensive compounds, and tissue hardness may affect the suitability of a host plant for the insect[6]. For example, fruits with tougher tissues or higher concentrations of defensive compounds may reduce larval feeding success, while softer tissues may allow easier penetration by the larvae. In macadamia, for instance, higher dry matter content and certain defensive compounds in the fruit husk have been associated with lower larval survival rates[6]. These plant insect interactions play an important role in determining the severity of infestations in different crop varieties. Because of its broad host range and concealed feeding behavior the CFB can persist in diverse agricultural environments and can move between different fruit crops within the same region. This flexibility in host plant use increases the difficulty of controlling the pest and highlights the importance of monitoring programs and integrated pest management strategies in affected agricultural systems.

IV. Agricultural Impact and Economic Importance

The CFB is considered one of the most economically important pests affecting citrus production in several regions of Central and South America. The damage caused by this insect results primarily from the feeding activity of the larval stage, which develops inside the fruit and consumes the pulp or seeds. Because the larvae feed internally, infested fruit often becomes unsuitable for fresh consumption or industrial processing and may drop prematurely from the tree[3]. This hidden feeding behavior makes early detection difficult and allows infestations to spread within orchards before growers are aware of the problem.
Significant crop losses attributed to the citrus fruit borer have been documented in major citrus producing regions, particularly in Brazil. Studies evaluating fruit drop in commercial orange orchards have shown that pests and diseases together can cause substantial losses in citrus production, with fruit borers and fruit flies accounting for a large proportion of these losses[3]. In some citrus growing regions of São Paulo state, fruit borers and fruit flies were responsible for approximately 3.9% of total fruit drop over a five-season evaluation period,

representing nearly one quarter of the overall fruit loss observed in orchards[3]. These losses translate into millions of boxes of oranges that cannot be harvested or marketed.

Economic assessments of the pest's impact have estimated that the citrus fruit borer can cause losses equivalent to tens of millions of dollars annually in affected regions[8]. In Brazil alone, damage caused by CFB has been estimated to result in the loss of approximately 30 million boxes of oranges each year, corresponding to roughly 80 million dollars in economic damage[8]. These losses are comparable to those caused by other major citrus pests and diseases, demonstrating the significant threat that the citrus fruit borer poses to commercial fruit production.

Beyond citrus orchards, the insect also affects several other fruit crops including macadamia, cacao, and guava, where larvae bore into fruits or nuts and feed on internal tissues[7]. In macadamia production systems, infestations of the citrus fruit borer have been reported to damage a substantial percentage of harvested nuts in some regions, further illustrating the pest's ability to affect multiple agricultural commodities[6]. The broad host range of the insect increases the difficulty of managing its populations because the pest can persist in alternative host plants even when citrus fruit availability is reduced.

As a result, researchers and agricultural agencies have devoted considerable effort to developing monitoring systems and integrated pest management strategies aimed at reducing crop losses and improving the sustainability of fruit production systems.

V. Management and Control Strategies

Managing populations of the CFB can be challenging due to the insect's concealed larval feeding behavior and its ability to utilize multiple host plants.. As a result, effective management of this pest typically relies on integrated pest management (IPM) strategies that combine several complementary control methods. These approaches may include the use of chemical insecticides, biological control agents such as parasitoid wasps, cultural practices that reduce pest habitats, and monitoring systems that detect pest populations early. By integrating multiple control strategies, growers can reduce crop losses while minimizing environmental impacts and decreasing reliance on broad spectrum pesticides[8][9].

Chemical Control

Chemical control remains one of the most widely used methods for managing populations of CFB particularly in large commercial citrus orchards. In many citrus producing regions, insecticides are applied during periods when adult moth activity and egg laying are expected to occur in order to reduce larval infestations before they penetrate the fruit[3]. Because the larvae quickly bore into the fruit after hatching, chemical treatments are most effective when applied during the early stages of infestation or when adult populations are detected through monitoring programs.

In Brazil and other major citrus producing areas, insecticide applications are commonly scheduled during critical periods of fruit development to prevent the establishment of larval populations within the fruit. However, despite regular insecticide treatments, the citrus fruit borer continues to cause substantial economic losses in citrus production systems[8]. This persistence highlights the challenges associated with controlling pests that develop internally within plant tissues, where they are protected from many chemical treatments once they enter the fruit.

Recent studies have also evaluated the effects of different insecticides on beneficial organisms used in biological control programs. Certain products, such as abamectin and cyantraniliprole, have been identified as relatively compatible with natural enemies used in integrated pest management programs[8]. These insecticides may have shorter persistence in the environment and allow beneficial parasitoids to be released shortly after application without significant harm to their populations.

While chemical control can help reduce pest populations, excessive reliance on insecticides may lead to several challenges, including increased production costs, environmental concerns, and potential impacts on beneficial insects that contribute to natural pest regulation. For this reason, modern pest management programs increasingly emphasize the integration of chemical control with other strategies such as biological control, monitoring systems, and cultural practices in order to achieve more sustainable pest management.

Biological Control

Biological control is an important part of managing CFB. Natural enemies, especially parasitoid wasps, can help reduce populations of this pest in agricultural systems. One group of parasitoids that has received significant attention is the genus *Trichogramma*, which is known for attacking the eggs of many moth and butterfly species.

One parasitoid species that has been studied for controlling the CFB is *Trichogramma atopovirilia* (an endoparasitoid). This tiny wasp lays its eggs inside the eggs of CFB preventing the pest from developing into larvae that would later damage fruit[8]. Laboratory and field studies have shown that releasing *T. atopovirilia* in orchards can reduce the number of citrus fruit borer eggs that successfully hatch, helping to lower pest populations before larvae begin feeding inside the fruit.

The success of biological control programs often depends on how well they work alongside other pest management practices, especially the use of insecticides. Some research has shown that certain insecticides are less harmful to parasitoid populations than others[8]. For example, products such as abamectin and cyantraniliprole have relatively short persistence and lower toxicity to parasitoids, allowing beneficial insects like *T. atopovirilia* to be released shortly after chemical treatments.

In addition to parasitoid wasps, other natural enemies such as predators and entomopathogenic microorganisms may also help regulate citrus fruit borer populations in orchards. However, heavy use of broad-spectrum insecticides can reduce populations of these beneficial organisms. For this reason, many modern pest management programs aim to combine biological control with selective insecticides and other management strategies to maintain natural enemies while still reducing pest populations.

Monitoring and Early Detection

Since infestations are often difficult to detect during early stages of damage, environmental monitoring plays a critical role in the effective management of the CFB. As a result, monitoring programs focus primarily on detecting adult moth populations before egg laying occurs. Early detection allows growers to implement control measures at the appropriate time and reduce the likelihood of large infestations.

One of the most commonly used monitoring tools for this pest is the pheromone-baited trap. These traps use synthetic versions of the sex pheromones produced by female moths to attract male moths into a trap where they can be counted[9]. The number of captured moths provides an estimate of the pest population within an orchard and helps growers determine when control measures may be necessary.

Pheromone traps are typically placed in citrus orchards before or during the fruit development period, when adult moth activity begins to increase. Monitoring the number of moths captured in these traps allows growers to track population trends over time and identify periods of increased pest activity. When trap counts reach certain thresholds, insecticide applications or other management strategies can be implemented to reduce pest populations before larvae enter the fruit.

In addition to pheromone traps, regular field inspections may also help detect early signs of infestation. One common indicator of citrus fruit borer activity is the presence of small entry holes in fruit accompanied by frass,

which is the waste material produced by feeding larvae[3]. Identifying these symptoms early can help growers remove infested fruit and prevent the spread of the pest within orchards.

By combining pheromone monitoring with regular field inspections, growers can improve the timing and effectiveness of pest management strategies while reducing unnecessary insecticide applications.

Field Sanitation and Cultural Control

Field sanitation and cultural management practices are important components of integrated pest management programs for CFB. Because larvae develop inside the fruit and later exit to pupate in the soil or nearby plant debris, removing infested fruit from orchards can significantly reduce the number of individuals that survive to adulthood[4]. By interrupting the pest's life cycle, sanitation practices can help lower overall pest populations in subsequent generations.

One common sanitation practice involves collecting and destroying fallen fruit from the ground beneath trees. Infested fruit that falls to the ground may still contain developing larvae that will eventually emerge and pupate in the soil. If these fruits are left in the orchard, they can serve as a source of new adult moths that will continue the infestation cycle. Removing and properly disposing of infested fruit can therefore reduce the number of pests emerging in future generations.

Pruning and general orchard maintenance may also contribute to pest management by improving airflow and reducing areas where pests can hide or develop. Maintaining orchard cleanliness by removing plant debris and fallen fruit can help eliminate potential pupation sites for the insect. In addition, reducing alternative host plants near orchards may limit the availability of additional food sources that allow the pest to persist in the surrounding environment.

Cultural control practices are most effective when used together with other management strategies such as monitoring programs, biological control, and carefully timed insecticide applications. By combining these approaches, growers can reduce pest populations while minimizing environmental impacts and promoting more sustainable agricultural production systems.

VI. Available Datasets and Resources

Several scientific databases and research tools provide valuable information for studying the CFB. These resources support research efforts related to pest identification, distribution, biology, and management. Access to reliable datasets and identification tools is essential for researchers, agricultural agencies, and growers who aim to monitor and control this pest effectively.

One important resource is the EPPO Global Database, which provides information on the taxonomy, geographic distribution, host plants, and regulatory status of numerous plant pests[2]. This database is widely used by plant protection organizations and researchers to track pest distribution and assess potential risks associated with the spread of agricultural pests.

Another important resource is the Cooperative Agricultural Pest Survey (CAPS) program maintained by the United States Department of Agriculture. The CAPS pest datasheets provide detailed information about pest biology, identification, monitoring methods, and management strategies[4]. These datasheets are designed to support early detection and rapid response efforts for economically important agricultural pests.

The USDA Tortricid Identification Tools database also provides detailed diagnostic information and images that assist with the identification of tortricid moth species[1]. These tools are particularly valuable for entomologists and plant protection specialists who need to accurately identify pest species during surveys and monitoring programs.

In addition to these databases, pheromone monitoring systems have become valuable tools for studying citrus fruit borer populations in agricultural environments[9]. These systems allow researchers and growers to monitor adult moth populations and track seasonal fluctuations in pest activity. Together, these datasets and research resources provide essential information that supports scientific research, pest monitoring programs, and the development of effective pest management strategies.

VII. Ongoing Research and Prospects

Ongoing research continues to focus on improving management strategies for the CFB through more sustainable and environmentally friendly approaches. In recent years, researchers have increasingly emphasized integrated pest management strategies that combine biological control, monitoring systems, and selective chemical treatments to reduce reliance on broad spectrum insecticides.

One area of active research involves the development and refinement of pheromone-based monitoring and control systems. Synthetic pheromone lures are already used to detect adult moth populations, but ongoing studies are exploring additional applications such as mating disruption and improved trap designs to enhance monitoring accuracy[9]. These technologies may help growers detect pest populations earlier and reduce unnecessary pesticide applications.

Biological control research also continues to investigate the use of parasitoid wasps such as *T. atopovirilia* to reduce citrus fruit borer populations[8]. Researchers are studying how biological control agents can be integrated more effectively with other pest management strategies while minimizing the negative impacts of insecticides on beneficial organisms.

Recent research programs in Brazil have also focused on developing comprehensive integrated pest management packages for citrus orchards. These programs combine monitoring systems, biological control agents, selective insecticides, and improved orchard management practices to create more effective and sustainable pest control strategies[10]. By integrating multiple control methods, researchers aim to reduce pesticide use while maintaining effective protection of citrus crops.

In addition to academic research, international collaborations between agricultural organizations and private industry are contributing to the development of sustainable pest management strategies. For example, projects supported by the Lindt Cocoa Foundation in partnership with the Alliance of Bioversity International and CIAT are exploring environmentally friendly pest management solutions in tropical agricultural systems. These initiatives include research on improved pest monitoring systems and pheromone-based control strategies aimed at reducing crop losses while supporting sustainable farming practices [11].

Future research will likely continue to focus on improving integrated pest management strategies that balance effective pest control with environmental sustainability. Advances in monitoring technologies, biological control methods, and ecological research will play an important role in helping farmers manage citrus fruit borer populations more effectively in the years ahead.

VIII. Conclusion

The CFB is an important agricultural pest that affects citrus and several other fruit crops throughout tropical and subtropical regions of the Americas. Its concealed larval feeding behavior and wide host range make it particularly difficult to manage, often resulting in significant economic losses in commercial fruit production systems. Because larvae develop inside fruit, infestations are frequently detected only after damage has already occurred, increasing the importance of preventive monitoring and effective management strategies.
Research on the biology, ecology, and host plant interactions of CFB has contributed greatly to the development of management approaches aimed at reducing crop damage. Integrated pest management programs that combine

monitoring systems, biological control agents, selective insecticides, and cultural practices have proven to be the most effective strategies for controlling this pest while minimizing environmental impacts. Continued research on pheromone monitoring, biological control methods, and sustainable agricultural practices will remain important for improving pest management programs.

As global agricultural production continues to expand and diversify, maintaining effective control of pests such as the citrus fruit borer will remain a critical challenge for growers and researchers alike. Advances in pest monitoring technologies, ecological research, and collaborative international research efforts offer promising opportunities to improve the sustainability and effectiveness of pest management strategies in the future.

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